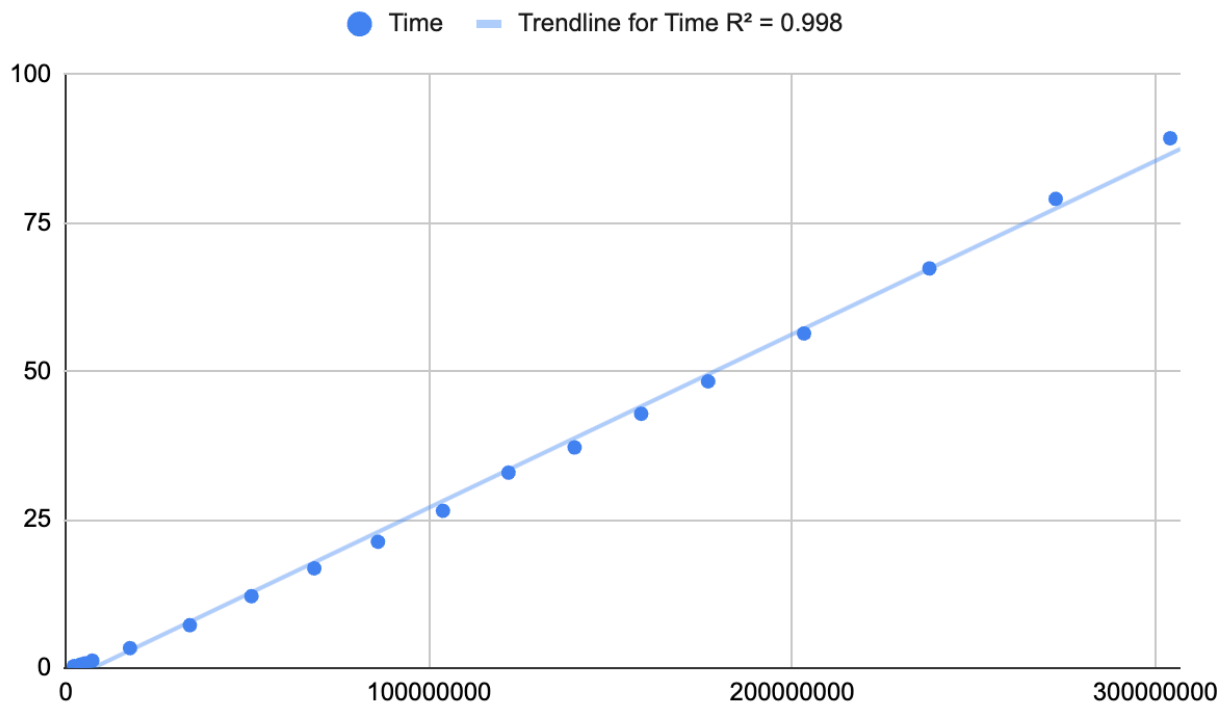


The purpose of this performance validation is to analyze the relationship between $n \log n$, with n as the number of inputs, and the corresponding running times, $Time$. The analysis aims to validate the expected time complexity of an algorithm. This report presents the data analyzed, the methodology used for linear regression, and discusses the challenges encountered during the analysis. The following is the table containing the data analyzed:

n	$n \log n$	$Time$
400000	2240823.997	0.327017
700000	4091568.628	0.590203
900000	5358818.258	0.78991
1200000	7295017.495	1.247234
2746666	17685249.11	3.357084
5093333	34160994.98	7.218737
7440000	51124502.64	12.110833
9786666	68415007.59	16.798695
12133333	85952289.72	21.282562
14500000	103839836	26.497394
16866666	121895888.4	32.932539
19233333	140096646.8	37.182287
21600000	158424201	42.847598
23966666	176864591.1	48.307469
27333333	203269595	56.381778
31700000	237783578.6	67.341756
36066666	272559618	79.066695
40000000	304082399.7	89.282916

To analyze the relationship between n , and $Time$, I performed a linear regression on the variables. Firstly, $n \log n$ was calculated based on individual n values. Then, a linear regression model was fitted with $n \log n$ as the independent variable and $Time$ as the dependent variable. The coefficient of determination, R^2 , was calculated to assess the fit of the regression model. As shown by the graph below, $R^2=0.998$.

I performed the linear regression by loading the dataset values into Google Sheets, then using the chart function to plot the data points on a graph, with $n \log n$ as the x-axis and $Time$ as the y-axis. I added a trendline to show the relationship between $n \log n$ and $Time$, then used Google's Sheet's in-built R^2 button for these types of charts to reveal the R^2 value.



Achieving a high R^2 value indicates that the linear regression model explains a large proportion of the variability in the dependent variable (*Time*) based on the independent variable ($n \log n$). A high R^2 value suggests that the model accurately captures the relationship between the input size n , the derived quantity related to it ($n \log n$), and the running time (*Time*).

There were some challenges encountered during the analysis. The R^2 was below 0.9 on a few trials, but after reducing the number of tasks the computer was performing and connecting the computer to a charger, the R^2 value stabilized to be above 0.9.

In conclusion, the performance validation analysis confirms the expected time complexity of the algorithm by demonstrating a strong linear relationship between $n \log n$ and *Time*. The regression model provides a reliable estimate of the running time for different numbers of inputs.